



SYDNEY GRAMMAR SCHOOL
MATHEMATICS DEPARTMENT
HALF-YEARLY EXAMINATIONS 2005

FORM VI

MATHEMATICS EXTENSION 1

Examination date

Wednesday 18th May 2005

Time allowed

2 hours

Instructions

- All seven questions may be attempted.
- All seven questions are of equal value.
- All necessary working must be shown.
- Marks may not be awarded for careless or badly arranged work.
- Approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

Collection

- Write your candidate number clearly on each booklet.
- Hand in the seven questions in a single well-ordered pile.
- Hand in a booklet for each question, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Keep the printed examination paper and bring it to your next Mathematics lesson.

Checklist

- SGS booklets: 7 per boy. A total of 1000 booklets should be sufficient.
- Candidature: 122 boys.

Examiner

KWM

QUESTION ONE (12 marks) Use a separate writing booklet.**Marks**

(a) When the polynomial $P(x) = kx^3 + x^2 - (2k - 1)x + 2$ is divided by $(x + 1)$, the remainder is 4. Find the value of k . **1**

(b) Differentiate the following with respect to x .

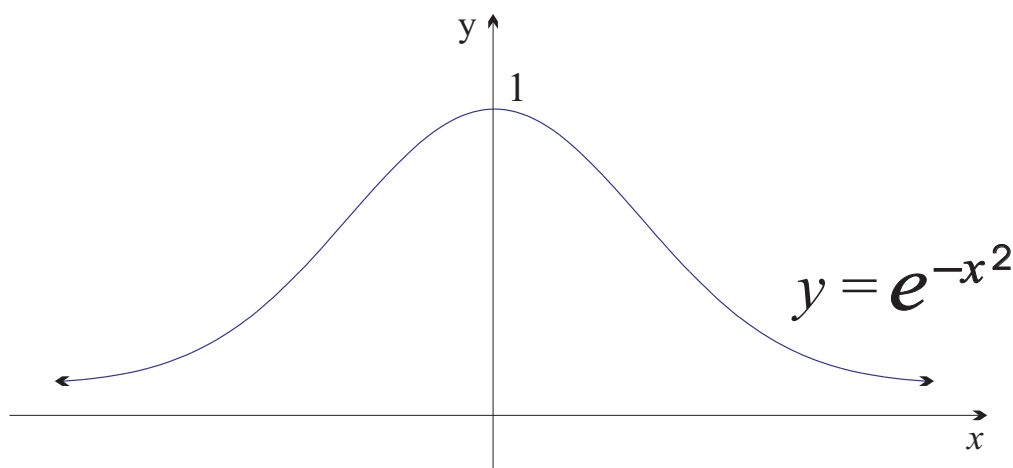
(i) $y = e^x \log_e x$ **1**

(ii) $y = \sin^{-1} 2x$ **2**

(c) Given that $\tan \alpha = \frac{1}{4}$ and $\tan \beta = \frac{3}{5}$, find the exact value of $\tan(\alpha + \beta)$. **2**

(d) Given that $x(2x - 1)(x + 1) + 3 = 2x^3 + bx^2 + cx + 3$, for all values of x , find the values of b and c . **2**

(e)



The diagram above shows the curve $y = e^{-x^2}$.

(i) Find $\frac{dy}{dx}$. **1**

(ii) Show that $\frac{d^2y}{dx^2} = 2e^{-x^2}(2x^2 - 1)$. **1**

(iii) For what values of x is the curve concave down? **2**

QUESTION TWO (12 marks) Use a separate writing booklet.

Marks

- (a) The displacement x metres of a particle moving in a straight line is given by

$$x = 5 + 4t - t^2$$

where t is the time in seconds.

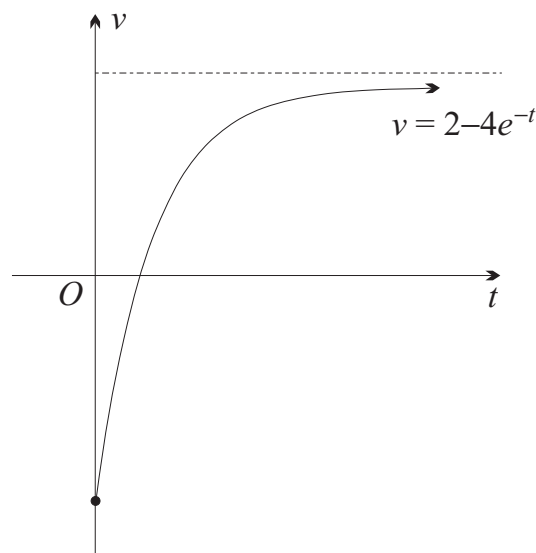
- | | |
|---|---|
| (i) Where is the particle initially? | 1 |
| (ii) Where does the particle change direction? | 2 |
| (iii) Find the distance travelled by the particle during the first 6 seconds. | 1 |
- (b)
- | | |
|---|---|
| (i) Without using calculus, sketch the polynomial function $y = x^3(x - 1)^2$. | 2 |
| (ii) Solve the inequation $x^3(x - 1)^2 > 0$. | 1 |
- (c) Calculate the area of the region bounded by the curve $y = \frac{1}{x^2 + 1}$ and the x -axis, between $x = 0$ and $x = \sqrt{3}$. 2
- (d) Find the following indefinite integrals.
- | | |
|-----------------------------------|---|
| (i) $\int \sec^2 3x \, dx$ | 1 |
| (ii) $\int \sin^3 x \cos x \, dx$ | 2 |

QUESTION THREE (12 marks) Use a separate writing booklet.**Marks**

(a) Solve $\cos^2 x - \sin^2 x = \frac{\sqrt{3}}{2}$, for $0 \leq x \leq 2\pi$.

3

(b)



The graph above shows the velocity $v = 2 - 4e^{-t}$ m/s at time t seconds of a particle moving in a straight line.

(i) Find the initial velocity of the particle.

1(ii) Show that the particle comes to rest at $t = \ln 2$ seconds.**1**

(iii) Calculate the exact distance travelled by the particle before it comes to rest.

2

(iv) To what value does the velocity of the particle approach as time increases?

1(c) (i) Given that $f(x) = \ln \left(\frac{1 + \sin x}{\cos x} \right)$, use the laws of logarithms to show that**2**

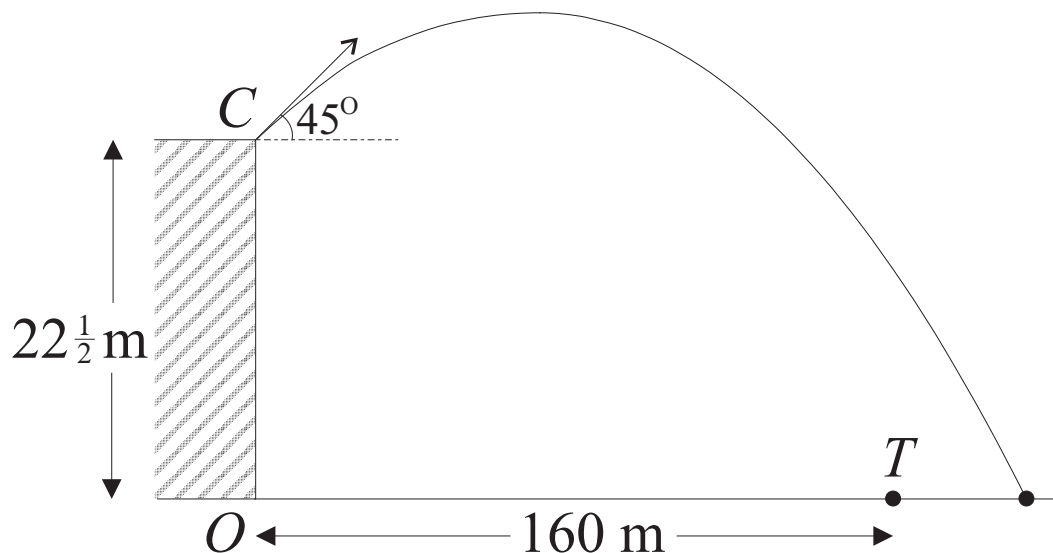
$$f'(x) = \sec x.$$

(ii) Hence or otherwise find $\int_0^{\frac{\pi}{4}} \sec x \, dx$.**2**

QUESTION FOUR (12 marks) Use a separate writing booklet.

Marks

- (a) Given that α , β and γ are the roots of the equation $x^3 - 4x^2 + 3x - 1 = 0$, find the value of :
- (i) $\alpha + \beta + \gamma$ 1
 - (ii) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ 1
 - (iii) $\alpha^2 + \beta^2 + \gamma^2$ 2
- (b) A particle is moving in simple harmonic motion about the origin on the x -axis. Its displacement in centimetres from the origin at any time t seconds is given by
- $$x = 6 \cos 2t.$$
- (i) Calculate the maximum velocity of the particle. 1
 - (ii) Find the first time the acceleration of the particle is zero. 2
 - (iii) Express the acceleration as a function of the displacement. 1
- (c) Consider the curve $y = \sin^{-1} x$.
- (i) Sketch the curve showing all relevant features. 2
 - (ii) Express x as a function of y , giving any restrictions on y . Hence calculate the area of the region bounded by the curve, the x -axis and the line $x = 1$. 2

QUESTION FIVE (12 marks) Use a separate writing booklet.**Marks**

The diagram above shows a cannon C positioned on the edge of a vertical cliff of height $22\frac{1}{2}$ metres. The cannon fires a shell with a muzzle velocity of 40 m/s. Set the origin O at the base of the cliff and use $g = 10$ m/s².

- (a) Suppose that the cannon fires a shell at an angle of elevation of 45° .

(i) Show that the shell's trajectory is defined by the equations

3

$$x = 20t\sqrt{2}$$

and $y = 20t\sqrt{2} - 5t^2 + 22\frac{1}{2}.$

(ii) Show that the shell hits the ground 180 metres from the foot of the cliff.

2

(iii) Calculate the maximum height above the ground reached by the shell.

2

(iv) Given that θ is the acute angle at which the shell strikes the ground, show that $\tan \theta = \frac{5}{4}.$

2

- (b) Suppose now that the angle of elevation is altered. Calculate the smaller angle of projection α , correct to the nearest minute, required for the shell to hit a target T positioned on the ground 160 metres from the base of the cliff. You may use the equations

3

$$x = 40t \cos \alpha$$

and $y = 40t \sin \alpha - 5t^2 + 22\frac{1}{2}.$

QUESTION SIX (12 marks) Use a separate writing booklet.**Marks**

- (a) The velocity
- v
- of a particle moving in a straight line is given by

$$v^2 = 6 + 10x - 4x^2,$$

where x is the displacement.

- (i) Find the acceleration as a function of the displacement, and hence show that the motion is simple harmonic.
- 1

- (ii) (
- α
-) Find the centre of motion.
- 1

- (
- β
-) Find the period of the motion.
- 1

- (
- γ
-) Find the amplitude of the motion.
- 2

- (b) Given that
- $\int_0^k \frac{6}{\sqrt{25-9x^2}} dx = \frac{\pi}{3}$
- , find the value of
- k
- .
- 3

- (c) Use the substitution
- $t = \tan \frac{\theta}{2}$
- to find the general solution of the equation
- 4

$$\cos \theta + \tan \frac{\theta}{2} - 1 = 0.$$

QUESTION SEVEN (12 marks) Use a separate writing booklet.**Marks**

- (a) When a polynomial
- $P(x)$
- is divided by
- $(x-1)$
- , the remainder is 3. When
- $P(x)$
- is divided by
- $(x+2)$
- , the remainder is
- -2
- . Find the remainder when the polynomial is divided by
- $x^2 + x - 2$
- .
- 2

- (b) The tangent to the curve
- $y = x^3 - 4x^2 - x + 2$
- , at a point
- Q
- on the curve, intersects the curve again at
- $A(2, -8)$
- . Find the co-ordinates of the point
- Q
- .
- 3

- (c) Rationalise the numerator of the expression
- $\sqrt{\frac{1+x}{1-x}}$
- and hence find the indefinite
- 3

integral $\int \sqrt{\frac{1+x}{1-x}} dx.$

- (d) (i) Use the definition of the derivative
- $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$
- to find the
- 2

derivative of $f(x) = e^x$ at $x = 0$, and hence show that $\lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1$.

- (ii) Find
- $\lim_{n \rightarrow \infty} \frac{e^{\frac{1}{n}} + e^{\frac{2}{n}} + e^{\frac{3}{n}} + \dots + e^{\frac{n}{n}}}{n}$
- .
- 2

END OF EXAMINATION

QUESTION 1

(a) $P(x) = Kx^3 + x^2 - (2K-1)x + 2$

$P(-1) = 4$ (Using the remainder theorem.)

$$-K + 1 + (2K-1) + 2 = 4$$

$$K + 2 = 4 \quad \checkmark$$

$$K = 2$$

(b)

(i) $y = e^x \ln x$

$$\frac{dy}{dx} = e^x \ln x + e^x \times \frac{1}{x} \quad \checkmark$$

$$\frac{dy}{dx} = e^x \left(\ln x + \frac{1}{x} \right)$$

(ii) $y = \sin^{-1} 2x$

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-4x^2}} \times 2 \quad \checkmark$$

$$= \frac{2}{\sqrt{1-4x^2}}$$

(c) $\tan \alpha = \frac{1}{4}$ and $\tan \beta = \frac{3}{5}$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$= \frac{\frac{1}{4} + \frac{3}{5}}{1 - \frac{3}{20}} \quad \checkmark$$

$$= \frac{5+12}{20-3}$$

$$= 1 \quad \checkmark$$

(d)

$$3 + x(2x-1)(x+1) \equiv 2x^3 + bx^2 + cx + 3$$

$$x(2x^2 + x - 1) \equiv 2x^3 + bx^2 + cx + 3$$

$$2x^3 + x^2 - x + 3 \equiv 2x^3 + bx^2 + cx + 3 \quad \checkmark$$

equating co-efficients:

$$b = 1 \quad \text{and} \quad c = -1 \quad \checkmark$$

(e) (i) $y = e^{-x^2}$

$$\frac{dy}{dx} = -2x e^{-x^2} \quad \checkmark$$

(ii) $\frac{d^2y}{dx^2} = -2e^{-x^2} + -2xe^{-x^2} \times -2x$

$$= -2e^{-x^2} + 4x^2 e^{-x^2} \quad \checkmark$$

$$= 2e^{-x^2} (2x^2 - 1)$$

(iii) Find the x -coordinates of the points of inflexion.

$$2e^{-x^2} (2x^2 - 1) = 0 \quad \checkmark$$

$$2x^2 = 1$$

$$x^2 = \frac{1}{2}$$

$$x = \pm \frac{1}{\sqrt{2}}$$

The curve is concave

$$\text{down} \quad -\frac{1}{\sqrt{2}} < x < \frac{1}{\sqrt{2}} \quad \checkmark$$

(from the graph.)

OR.

$$\text{Solve } 2e^{-x^2} (2x^2 - 1) < 0$$

$$2x^2 - 1 < 0$$

$$x^2 < \frac{1}{2}$$

$$-\frac{1}{\sqrt{2}} < x < \frac{1}{\sqrt{2}}$$

(12)

QUESTION 2

(a) $x = 5 + 4t - t^2$

(i) when $t=0$, $x=5$ ✓

(ii) $x = 5 + 4t - t^2$

$\dot{x} = 4 - 2t$

$4 - 2t = 0$

$2t = 4$

$t = 2\text{ s.}$ ✓

The particle changes direction

at $x = 5 + 8 - 4$

$x = 9.$ ✓

(iii) when $t=0$, $x=5$

when $t=2$, $x=9$

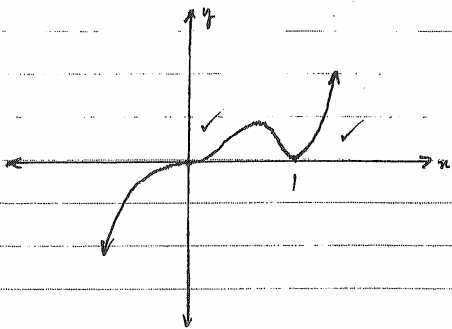
when $t=6$, $x=-7$

distance = $5 + 9 + 7$

travelled = 20 m. ✓

(b) (i) $y = x^3(x-1)^2$

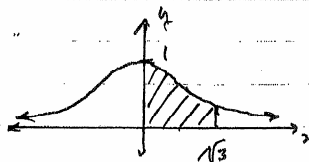
x intercepts at $x=0$ and $x=1$



(ii) $x^3(x-1)^2 > 0$

$x > 0, x \neq 1.$ ✓

(c)



$A = \int_0^{\sqrt{3}} \frac{dx}{1+x^2}$ ✓

$= \left[\tan^{-1}x \right]_0^{\sqrt{3}}$

$= \tan^{-1}\sqrt{3} - \tan^{-1}0$

$= \frac{\pi}{3}$ square units. ✓

(d)

(i) $\int \sec^2 3x \, dx = \frac{1}{3} \tan 3x + c$ ✓

(ii) $\int \sin^3 x \cos x \, dx = \frac{1}{4} \sin^4 x + c$ ✓

(12)

QUESTION 3

(a) $\cos^2 x - \sin^2 x = \frac{\sqrt{3}}{2}$

for $0 \leq x \leq 2\pi$.

$\cos 2x = \frac{\sqrt{3}}{2}$ ✓

✓ $2x = \frac{\pi}{6}, \frac{11\pi}{6}, \frac{13\pi}{6}, \frac{23\pi}{6}$

✓ $x = \frac{\pi}{12}, \frac{11\pi}{12}, \frac{13\pi}{12}, \frac{23\pi}{12}$

(b) (i) $v = 2 - 4e^{-t}$

when $t=0$, $v = 2-4$

$v = -2 \text{ m/s.}$ ✓

(ii) when $t = \ln 2$

$v = 2 - 4e^{-\ln 2}$

$v = 2 - 4e^{\ln \frac{1}{2}}$ ✓

$v = 2 - 4 \times \frac{1}{2}$

$v = 0$.

(iii) $\int_0^{\ln 2} 2 - 4e^{-t} dt$

$= \left[2t + 4e^{-t} \right]_0^{\ln 2}$ ✓

$= (2\ln 2 + 4e^{-\ln 2}) - (0 + 4)$

$= 2\ln 2 + 4 \times \frac{1}{2} - 4$ ✓

$= |\ln 4 - 2| \text{ m}$

(iv) as $t \rightarrow \infty$, $e^{-t} \rightarrow 0$

$v = 2 - 4e^{-t}$

$v \rightarrow 2 \text{ m/s.}$ ✓

(c)

(i) $f(x) = \ln \frac{(1 + \sin x)}{\cos x}$

$f(x) = \ln(1 + \sin x) - \ln \cos x$

$f'(x) = \frac{\cos x}{1 + \sin x} + \frac{\sin x}{\cos x}$ ✓

$= \frac{\cos^2 x + \sin x(1 + \sin x)}{\cos x(1 + \sin x)}$

$\cos x(1 + \sin x)$

$= \frac{\cos^2 x + \sin^2 x + \sin x}{\cos x(1 + \sin x)}$

$\cos x(1 + \sin x)$

$= \frac{1 + \sin x}{\cos x(1 + \sin x)}$ ✓

$\cos x(1 + \sin x)$

$= \frac{1}{\cos x}$

$f'(x) = \sec x$

(ii) $\int_0^{\frac{\pi}{4}} \sec x dx = \left[\ln \frac{(1 + \sin x)}{\cos x} \right]_0^{\frac{\pi}{4}}$ ✓

$= \ln \left(\frac{1 + \frac{1}{\sqrt{2}}}{\frac{1}{\sqrt{2}}} \right) - \ln 1$ ✓

$= \ln(\sqrt{2} + 1)$

(12)

QUESTION 4

(a) $x^3 - 4x^2 + 3x - 1 = 0$

(i) $\alpha + \beta + \gamma = -\frac{b}{a}$
 $= 4 \quad \checkmark$

(ii) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma}$

$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$
 $= 3$

$\alpha\beta\gamma = -\frac{d}{a}$
 $= 1$

$\therefore \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{3}{1} \quad \checkmark$

(iii) $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$
 $= 4^2 - 2 \times 3$
 $= 10 \quad \checkmark$

(b) (i) $x = 6 \cos 2t$

$\dot{x} = -12 \sin 2t$

Maximum velocity $|\dot{x}| = 12 \text{ m/s} \quad \checkmark$

(ii) $\ddot{x} = -24 \cos 2t$

So $\ddot{x} = 0$ when

$-24 \cos 2t = 0 \quad \checkmark$

at $2t = \frac{\pi}{2}$

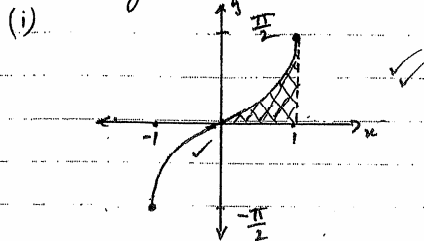
$t = \frac{\pi}{4} \text{ s} \quad \checkmark$

(iii) $\ddot{x} = -24 \cos 2t$

$= -4 (6 \cos 2t)$

$= -4x \quad \checkmark$

(c) $y = \sin^{-1}x$



(ii) $y = \sin^{-1}x$
 $x = \sin y, \quad -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$

Area shaded = Rectangle - area bounded by the curve and the y-axis.

$A = 1 \times \frac{\pi}{2} - \int_0^{\frac{\pi}{2}} \sin y \, dy \quad \checkmark$

$= \frac{\pi}{2} + \left[\cos y \right]_0^{\frac{\pi}{2}}$

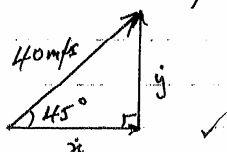
$= \frac{\pi}{2} + (0 - 1)$

$= \frac{\pi}{2} - 1 \text{ square units.} \quad \checkmark$

(12)

QUESTION 5

(i) initial components of velocity:



$$\dot{x} = 40 \cos 45^\circ$$

$$\dot{x} = 20\sqrt{2} \text{ m/s}$$

$$\dot{y} = 40 \sin 45^\circ$$

$$\dot{y} = 20\sqrt{2} \text{ m/s}$$

$$\ddot{x} = 0$$

$$\dot{x} = c_1$$

$$\ddot{y} = -10$$

$$\dot{y} = -10t + c_2$$

when $t=0$, $\dot{x} = 20\sqrt{2}$ and $\dot{y} = 20\sqrt{2}$.

$$c_1 = c_2 = 20\sqrt{2}$$

$$\dot{x} = 20\sqrt{2}$$

$$x = \int 20\sqrt{2} dt$$

$$\dot{y} = -10t + 20\sqrt{2}$$

$$y = \int -10t + 20\sqrt{2} dt$$

$$x = 20t\sqrt{2} + c_3$$

✓

$$y = -5t^2 + 20t\sqrt{2} + c_4$$

when $t=0$, $x=0$ and $y = 22\frac{1}{2}$

$$c_3 = 0$$

$$c_4 = 22\frac{1}{2}$$

$$x = 20t\sqrt{2}$$

✓

$$y = -5t^2 + 20t\sqrt{2} + 22\frac{1}{2}$$

are the equations of motion for the shell.

(ii) Put $x = 180$ m.

$$20t\sqrt{2} = 180$$

$$t = \frac{9\sqrt{2}}{2} \text{ s} \quad \checkmark$$

when $x = 180$,

$$y = -5t^2 + 20t\sqrt{2} + 22\frac{1}{2}$$

$$= -5 \times \frac{81 \times 2}{4} + 20 \times \frac{9\sqrt{2}}{2} \times \sqrt{2} + 22\frac{1}{2}$$

$$= -\frac{405}{2} + 180 + 22\frac{1}{2}$$

$$= -202\frac{1}{2} + 202\frac{1}{2}$$

$$= 0 \quad \checkmark$$

when $x = 180$, $y = 0$. The shell hits the ground 180m from the base of the cliff.

(iii) The maximum height is

reached when $\dot{y} = 0$.

$$-10t + 20\sqrt{2} = 0$$

$$t = 2\sqrt{2} \text{ s.} \quad \checkmark$$

$$y = -5t^2 + 20t\sqrt{2} + 22\frac{1}{2}$$

$$y = -5 \times 8 + 80 + 22\frac{1}{2}$$

$$y = 62\frac{1}{2} \text{ m} \quad \checkmark$$

The maximum height above the ground reached by the shell is $62\frac{1}{2}$ m.

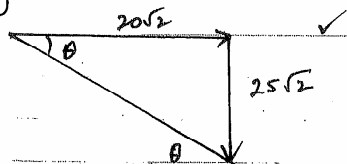
(iv) The shell strikes the ground
at $t = \frac{9\sqrt{2}}{2}$ s. (part (i)).

Components of velocity:

$$u_x = 20\sqrt{2}$$

$$u_y = -10 \times \frac{9\sqrt{2}}{2} + 20\sqrt{2}$$

$$u_y = -25\sqrt{2}$$



$$\tan \theta = \frac{25\sqrt{2}}{20\sqrt{2}}$$

$$\tan \theta = \frac{5}{4} \quad \checkmark$$

(b) The equations of motion are:

$$x = 40t \cos \alpha \quad \text{and} \quad y = 40t \sin \alpha - 5t^2 + 22\frac{1}{2}$$

put $x = 160$ and $y = 0$:

$$40t \cos \alpha = 160$$

$$t = \frac{4}{\cos \alpha} \quad \text{substitute.} \quad \checkmark$$

$$40t \sin \alpha - 5t^2 + 22\frac{1}{2} = 0$$

$$160 \tan \alpha - \frac{80}{\cos^2 \alpha} + 22.5 = 0$$

$$160 \tan \alpha - 80 \sec^2 \alpha + 22.5 = 0$$

$$160 \tan \alpha - (1 + \tan^2 \alpha) 80 + 22.5 = 0$$

$$-80 \tan^2 \alpha + 160 \tan \alpha - 80 + 22.5 = 0$$

$$80 \tan^2 \alpha - 160 \tan \alpha + 57.5 = 0 \quad \checkmark$$

using the quadratic formula

$$\tan \alpha = \frac{160 \pm \sqrt{160^2 - 4 \times 80 \times 57.5}}{160}$$

$$\tan \alpha = \frac{160 \pm \sqrt{7200}}{160}$$

$$\tan \alpha = \frac{160 - 60\sqrt{2}}{160}$$

$$\alpha = 25^\circ 9' \quad \checkmark$$

(12)

QUESTION 6

(a) $u^2 = 6 + 10x - 4x^2$

(i) $\frac{1}{2}u^2 = 3 + 5x - 2x^2$

$\frac{d}{dx} \left(\frac{1}{2}u^2 \right) = 5 - 4x \quad \checkmark$

$\ddot{x} = -4 \left(x - \frac{5}{4} \right)$

in x , the motion is SHM.

(ii) (a) centre of motion is $x = \frac{5}{4}$. $\checkmark$

(b) $n = 2$. $T = \frac{2\pi}{n}$

$T = \frac{\pi}{2} \text{ s} \quad \checkmark$

(c) when $v = 0$.

$6 + 10x - 4x^2 = 0$

$2x^2 - 5x - 6 = 0$

$(2x + 1)(x - 3) = 0$

$x = -\frac{1}{2} \quad \text{or} \quad x = 3 \quad \checkmark$



amplitude $= 3 - \frac{5}{4}$
 $= \frac{7}{4} \quad \checkmark$

(b) $\int_0^K \frac{6}{\sqrt{25 - 9x^2}} dx = \frac{\pi}{3}$

$6 \left[\frac{1}{3} \sin^{-1} \frac{3x}{5} \right]_0^K = \frac{\pi}{3} \quad \checkmark$

$\left[\sin^{-1} \frac{3x}{5} \right]_0^K = \frac{\pi}{6}$

$\sin^{-1} \frac{3K}{5} = \frac{\pi}{6}$

$\frac{3K}{5} = \frac{1}{2}$

$6K = 5$

$K = \frac{5}{6} \quad \checkmark$

(c) $t = \tan \frac{\theta}{2}$

$\cos \theta = \frac{1 - t^2}{1 + t^2}$

$\cos \theta + \tan \frac{\theta}{2} - 1 = 0$

$\frac{1 - t^2}{1 + t^2} + t - 1 = 0 \quad \checkmark$

$1 - t^2 + t(1 + t^2) - (1 + t^2) = 0$

$1 - t^2 + t + t^3 - 1 - t^2 = 0$

$t^3 - 2t^2 + t = 0$

$t(t^2 - 2t + 1) = 0$

$t(t - 1)(t - 1) = 0$

$t = 0 \quad \text{or} \quad t = 1 \quad \checkmark$

$\tan \frac{\theta}{2} = 0 \quad \text{or} \quad \tan \frac{\theta}{2} = 1$

$\frac{\theta}{2} = n\pi$

$\frac{\theta}{2} = \frac{\pi}{4} + \pi n$

$\theta = 2\pi n \quad \text{or} \quad \theta = \frac{\pi}{2} + 2\pi n$

(where n is an integer)

(12)

QUESTION 7.

(a) Using the division algorithm:

$$P(x) = (x^2 + x - 2)Q(x) + ax + b$$

$$P(x) = (x-1)(x-2)Q(x) + ax + b$$

$$P(1) = 3 \quad \text{and} \quad P(-2) = -2 \quad \checkmark$$

$$\textcircled{1} \quad a + b = 3$$

$$\textcircled{2} \quad -2a + b = -2$$

$$\textcircled{1} - \textcircled{2} \quad 3a = 5$$

$a = \frac{5}{3}$ and $b = \frac{4}{3}$, hence
the remainder is $\frac{5}{3}x + \frac{4}{3}$. $\checkmark$

(b) Let the tangent be

$y = mx + b$. Since it passes
through the point $(2, -8)$.

$$2m + b = -8$$

$$b = -2m - 8. \quad \checkmark$$

$$\textcircled{1} \quad y = mx - 8 - 2m$$

$$\textcircled{2} \quad y = x^3 - 4x^2 - x + 2.$$

Solving simultaneously:

$$x^3 - 4x^2 - (1+m)x + (10+2m) = 0. \quad \checkmark$$

The roots of this cubic α, α
and 2 correspond to the x
co-ordinates of the points of
intersection.

$$\text{Sum of roots: } \alpha + \alpha + 2 = 4$$

$$2\alpha = 2$$

$$\alpha = 1. \quad \checkmark$$

Therefore $O_1(1, -2)$. $\checkmark$

$$(c) \quad \sqrt{\frac{1+x}{1-x}} \times \sqrt{\frac{1+x}{1+x}} = \frac{1+x}{\sqrt{1-x^2}} \quad \checkmark$$

$$\begin{aligned} \int \sqrt{\frac{1+x}{1-x}} dx &= \int \frac{1+x}{\sqrt{1-x^2}} dx \\ &= \int \frac{1}{\sqrt{1-x^2}} + \frac{x}{\sqrt{1-x^2}} dx \\ &= \sin^{-1}x - \sqrt{1-x^2} + c \quad \checkmark \end{aligned}$$

$$(d)(i) \quad f(x) = e^x$$

at $x=0$, $f'(x) = e^x$, $f'(0) = 1$

Using the definition. $\checkmark$

$$f'(x) = \lim_{h \rightarrow 0} \frac{e^{x+h} - e^x}{h}$$

$$f'(0) = \lim_{h \rightarrow 0} \frac{e^h - 1}{h}$$

Therefore $\lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1 \dots (1) \quad \checkmark$

$$(ii) \quad \lim_{n \rightarrow \infty} \frac{e^{\frac{1}{n}} + e^{\frac{2}{n}} + e^{\frac{3}{n}} + \dots + e^{\frac{n}{n}}}{n}$$

The numerator is a GP.

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$$= e^{\frac{1}{n}} \left\{ \frac{(e^{\frac{1}{n}})^n - 1}{e^{\frac{1}{n}} - 1} \right\}$$

$$= (e-1) \frac{e^{\frac{1}{n}}}{e^{\frac{1}{n}} - 1}$$

$$\lim_{n \rightarrow \infty} \frac{e^{\frac{1}{n}} + e^{\frac{2}{n}} + e^{\frac{3}{n}} + \dots + e^{\frac{n}{n}}}{n}$$

$$= \lim_{n \rightarrow \infty} \frac{(e-1) e^{\frac{1}{n}}}{n(e^{\frac{1}{n}} - 1)} \quad \checkmark$$

(Now let $\frac{1}{n} = h$
as $n \rightarrow \infty$, $h \rightarrow 0$)

$$= \lim_{h \rightarrow 0} \frac{(e-1) e^h}{e^h - 1} \cdot h$$

$$= \lim_{h \rightarrow 0} (e-1) e^h \cdot \frac{h}{e^h - 1}$$

$$= (e-1) \lim_{h \rightarrow 0} \frac{h}{e^h - 1} \cdot e^h$$

from (1).

✓

$$= e-1.$$

(12)